

Assignment 2A - Dataset Cleaning and Exploration

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Dataset Overview

The uploaded dataset contains 36540 rows and 10 columns related to climate and energy information.

Cleaning Steps

- Checked all missing values.
- Filled numerical missing values using the median.
- Filled categorical missing values using the mode.
- Removed duplicate records.
- Standardized column names using lowercase and underscores.

Key Patterns

1. Numerical values show consistent trends across the dataset.
2. Some variables display noticeable variation between observations.
3. The dataset contains sufficient records for meaningful sustainability analysis.

Anomalies

1. A few observations have unusually high numerical values.
2. Some records lie outside the normal interquartile range.

Assumptions

Missing numerical values were replaced with the median because it is less sensitive to extreme values. Missing categorical values were replaced with the most frequent category. Column names were standardized only for consistency.

Conclusion

The cleaned dataset is consistent, organized and ready for visualization, predictive analysis and sustainability decision making.